

SQL - THE DIGITAL LIBRARY

Introduction to Relational Databases [cite: 304]

Planrs Academy | Grade 6 Session

Learn how to organize the world's data!

Why can't we use one giant notebook? [cite: 305, 306]

Every day, Indians create billions of pieces of data! [cite: 307] From school attendance to cricket scores, library books to train tickets - there's just too much information to keep in one notebook. [cite: 307]

The Problem: Imagine trying to find one specific student's grade in a notebook with a million pages. It would take forever! [cite: 307]

We need smart storage systems to organize it all! [cite: 308]

A Digital Library [cite: 310, 311]

A Relational Database is an organized collection of related tables that store information neatly. [cite: 311]

The Library Analogy: [cite: 312]

- **Tables = Shelves:** These organize the data by category. [cite: 313]
- **Rows = Books:** Each row is one single record (like one book). [cite: 313]
- **Columns = Book Info:** These are the details (title, author, pages). [cite: 313]

Just like bookshelves in your school library keep books organized by subject, databases keep our data tidy! [cite: 311]

Understanding the Structure [cite: 314, 315]

Think of a table like your class attendance register! [cite: 316]

- **Rows:** These go across. Each row is one student's record. [cite: 316, 319]
- **Columns:** These go down. They define the type of information (Roll No, Name, Date). [cite: 316, 319]
- **Cell:** Every little box holds just one single piece of information. [cite: 317, 319]

[Image of a labeled table structure showing rows, columns, and cells]

Our School Register is a Table! [cite: 323]

Every day, your teacher uses a relational table without even knowing it! [cite: 323]

Roll No	Name	Attendance
1	Aarav	Present
2	Diya	Absent
3	Ishaan	Present

Each row represents a unique student, and each column tells us a specific detail about them. [cite: 316]

Why do we need a Primary Key? [cite: 325]

Imagine your class has three students named 'Rahul'! How do we tell them apart? [cite: 326]

Primary Key = Your data's special ID card! [cite: 331]

A Primary Key is a unique piece of information that belongs only to one record. [cite: 327] It keeps data unique and prevents mix-ups. [cite: 328, 330]

Think of it like the **Aadhaar Number** or **Admission Number**. No two people in India have the same Aadhaar! [cite: 327, 392]

Admission Number is the Primary Key! [cite: 334]

Notice how we have multiple Rahuls, but each has a unique ID! [cite: 335]

Admission No (PK)	Name	Class
A101	Rahul Sharma	6A
A103	Rahul Verma	6A
A104	Rahul Singh	6C

Even if names repeat, the Primary Key ensures there is no confusion! [cite: 335, 388]

How Tables Talk to Each Other [cite: 369]

The "Relational" part of a database means tables can be linked! [cite: 369]

We use the **Student ID** to connect the Student Table to the Library Books Table. [cite: 369]

This lets us see exactly which books Priya (ID 001) or Meera (ID 003) have borrowed without writing their full names everywhere! [cite: 364-371]

Indian Examples [cite: 374, 375]

Databases are everywhere around us in India! [cite: 375]

- **Indian Railways:** Uses a database to manage millions of train tickets and PNR numbers. [cite: 378, 379]
- **Metro Cards:** Keeps track of your balance and where you traveled.
- **School Library:** Manages thousands of books and who borrowed them.

Each one stores information in tables with rows and columns. [cite: 376]

Design a Cricket Team Table [cite: 381]

Be a Database Architect! Design your own table with these columns: [cite: 382, 383]

- Player ID (Primary Key) [cite: 384]
- Name [cite: 385]
- Specialty (Batsman/Bowler) [cite: 386]
- Runs Scored [cite: 386]

Think: What makes each player unique? That's your Primary Key! [cite: 387]

Sample Cricket Team Table [cite: 388]

Player ID (PK)	Name	Specialty	Runs
P001	Virat	Batsman	1500
P002	Rohit	Opener	1200
P003	Jasprit	Bowler	50

Notice how IDs keep data organized even if we have players with similar names! [cite: 388, 389]

Why are Databases Important? [cite: 390]

Databases help us manage lots of information in neat tables—like a super-organized digital notebook! [cite: 390]

- **Organize Data Easily:** No more giant, messy notebooks! [cite: 391]
- **Primary Keys Keep Data Unique:** Like your Aadhaar number. [cite: 392]
- **Power Your Favorite Apps:** Databases run Zomato, Amazon, and Hotstar! [cite: 393]

Powering the Apps You Use [cite: 394]

Every time you do something online, a database is working behind the scenes! [cite: 395]

Zomato: Stores menus, prices, and your orders. [cite: 395]

IRCTC: Manages train seats and passenger lists. [cite: 395]

Hotstar: Remembers which movie you were watching! [cite: 393]

What did we learn today?

- ✓ Relational databases are organized collections of tables. [cite: 311]
- ✓ Tables consist of Rows (records) and Columns (info types). [cite: 319]
- ✓ Primary Keys are unique IDs that prevent confusion. [cite: 328]
- ✓ Databases connect tables to share information. [cite: 369]
- ✓ They power almost every app in your daily life! [cite: 393]

THANK YOU! [cite: 396]

You're now a Database Architect in training! [cite: 382]

Any Questions? [cite: 396]